

Report 3 - Logistic Regression

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1 Algorithm

In this report I use Logistic Regression to classify 2 class on a small dataset (50 points, 2 features X1 and X2). The model is:

$$\hat{y} = \sigma(w_1x_1 + w_2x_2 + w_0), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Loss function is Cross Entropy:

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

I use gradient descent to update the weights w_1, w_2, w_0 each epoch. The training stop when the loss change is smaller than the threshold.

Initial value: $w_1 = 1.0, w_2 = 1.0, w_0 = 0.0$. Learning rate = 0.001, threshold = 0.000001, max epochs = 100000.

Model converge at epoch 65931 with final loss ≈ 0.1158 and weights $w_1 = 0.6767, w_0 = -4.8414$.

2 Result

After training, I draw the decision boundary and also plot the loss curve to check the convergence.

The test point $x_1 = 6, x_2 = 4$ give the prediction $\hat{y} \approx 0.6683$, so it belong to class 1.

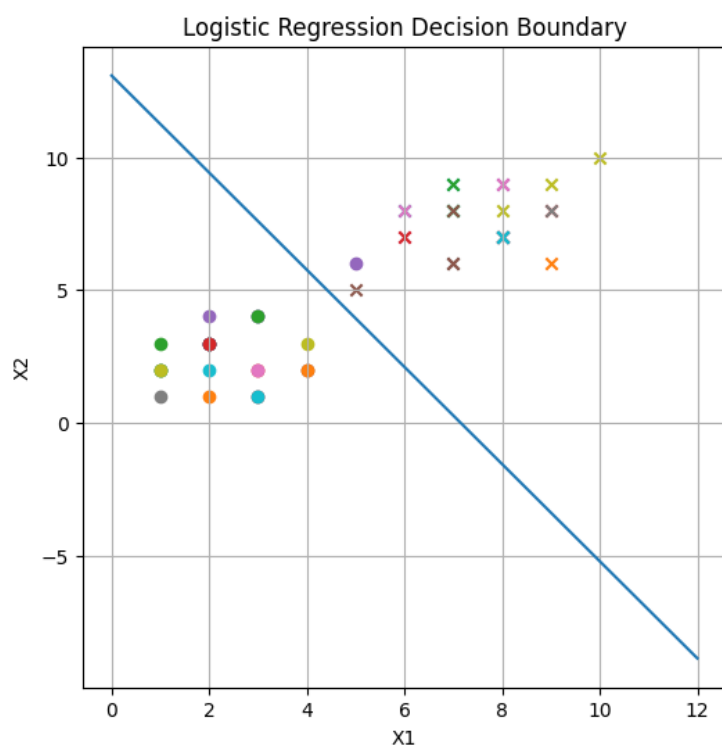


Figure 1: Decision boundary of the model

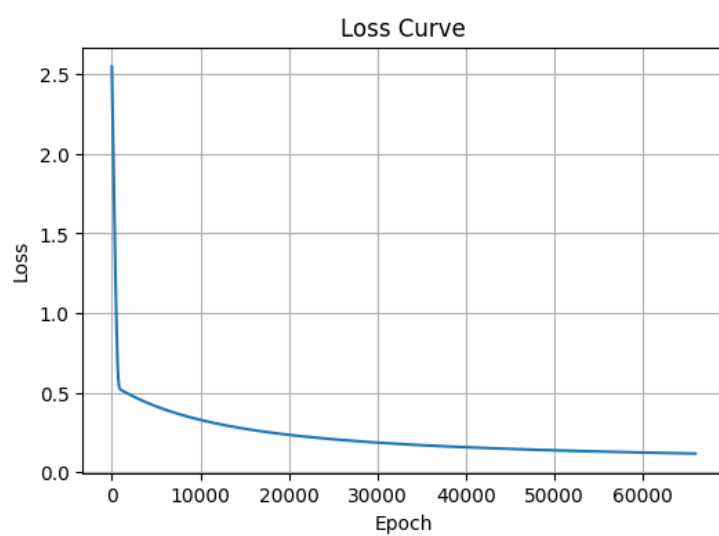


Figure 2: Loss of model over epochs